

# ETC5250 - Machine Learning Project

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## Summary of Visualisation

The data were strongly imbalanced, with far fewer deaths than survivors. Mortality rates varied across admission type, first care unit, ICD chapter and diagnosis group. Diagnosis and categorical importance plots showed that only a few variables had relatively large outcome differences. PCA, scatterplots and tour views showed substantial overlap between outcome groups, suggesting that mortality risk was spread across many weak signals rather than separated by a few variables.

## Preprocessing

I joined the predictors with the outcome and converted the response to a yes/no factor. I created ICD-based features, grouped rare diagnoses into DIAG\_OTHER, and summarised secondary diagnoses using counts and dummy flags. Missing diagnosis flags were set to 0. Identifiers and raw dates were removed, factor levels were aligned, categorical variables were dummy-coded, zero-variance predictors were removed, and numeric predictors were standardised for scale-sensitive models.

## kNN and Logistic Regression

I fitted kNN as a distance-based benchmark using an 80/20 validation split because tuning was slow in the high-dimensional dummy-coded data. The best kNN model used 150 neighbours with Manhattan distance and achieved ROC-AUC 0.787, suggesting that kNN was not suitable for this dataset.

I then fitted logistic regression models using 5-fold cross-validation. The best model was ridge-regularised logistic regression using the full feature set, with mixture = 0 and penalty 0.146. It achieved ROC-AUC 0.932. Since the full feature set performed well, later experiments focused more on model choice and hyperparameter tuning than further variable removal.

## Tree-based Models

I first fitted a regularised decision tree. The best single tree used tree\_depth = 15, min\_n = 30 and cost\_complexity = 1e-04, achieving ROC-AUC 0.862. This suggested that one tree was not flexible enough, so I moved to ensemble tree models.

Random forests improved performance substantially. During RF tuning, I compared different diagnosis feature sizes and found that keeping 75 diagnosis groups worked best among the settings I tested. I therefore used this feature set in the later boosted tree models as well. The best random forest used mtry = 22, trees = 1000 and min\_n = 1, with 75 diagnosis groups, 250 secondary ICD-3 flags and 200 secondary full ICD-9 flags. It achieved ROC-AUC 0.940. Further RF tuning produced only small improvements, so I then fitted boosted trees.

The best boosted tree used mtry = 60, trees = 1000, min\_n = 3, tree\_depth = 5, learn\_rate = 0.05, loss\_reduction = 0.01 and sample\_size = 0.8, using the same 75 diagnosis groups, 250 secondary ICD-3 flags and 200 secondary full ICD-9 flags. It achieved 5-fold CV ROC-AUC 0.942 and Kaggle public ROC-AUC 0.946, making it the strongest single model. Variable importance suggested that both vital signs and secondary diagnosis variables were important.

## Neural Network Models

I fitted neural networks by hand-tuning the architecture using an 80/20 validation split, since neural networks were slower to fit repeatedly. I started with a small network and gradually increased hidden nodes and layers. The best neural network used one hidden layer with 350 ReLU nodes, achieving validation ROC-AUC 0.918 and Kaggle public ROC-AUC 0.939. It did not outperform the boosted tree and was less competitive on the public leaderboard, so it was kept as a benchmark rather than included in the final model.

## Ensemble Models

I tested weighted averages of predicted probabilities from the strongest models. I tried boosted tree with random forest, boosted tree with ridge logistic regression, and boosted tree with both random forest and ridge logistic regression. I did not include kNN because its performance was much weaker, and I did not include the neural network because it was tuned using a separate validation split and did not improve over the strongest tree-based models. Among the ensemble combinations I tested, the best ensemble combined boosted tree, random forest and ridge logistic regression with weights 0.45/0.4/0.15. It achieved the highest 5-fold CV ROC-AUC (0.948) so it was selected as the final submission model.

## Generative AI Use

I used generative AI to improve grammar, debug code, write helper functions and organise scripts. Model choices, variable selection, interpretation and final decisions were made by me. The main modelling code followed lecture slides and tutorial examples, with adjustments for this dataset.